Elliptic curves: Algebraic formulae for addition.

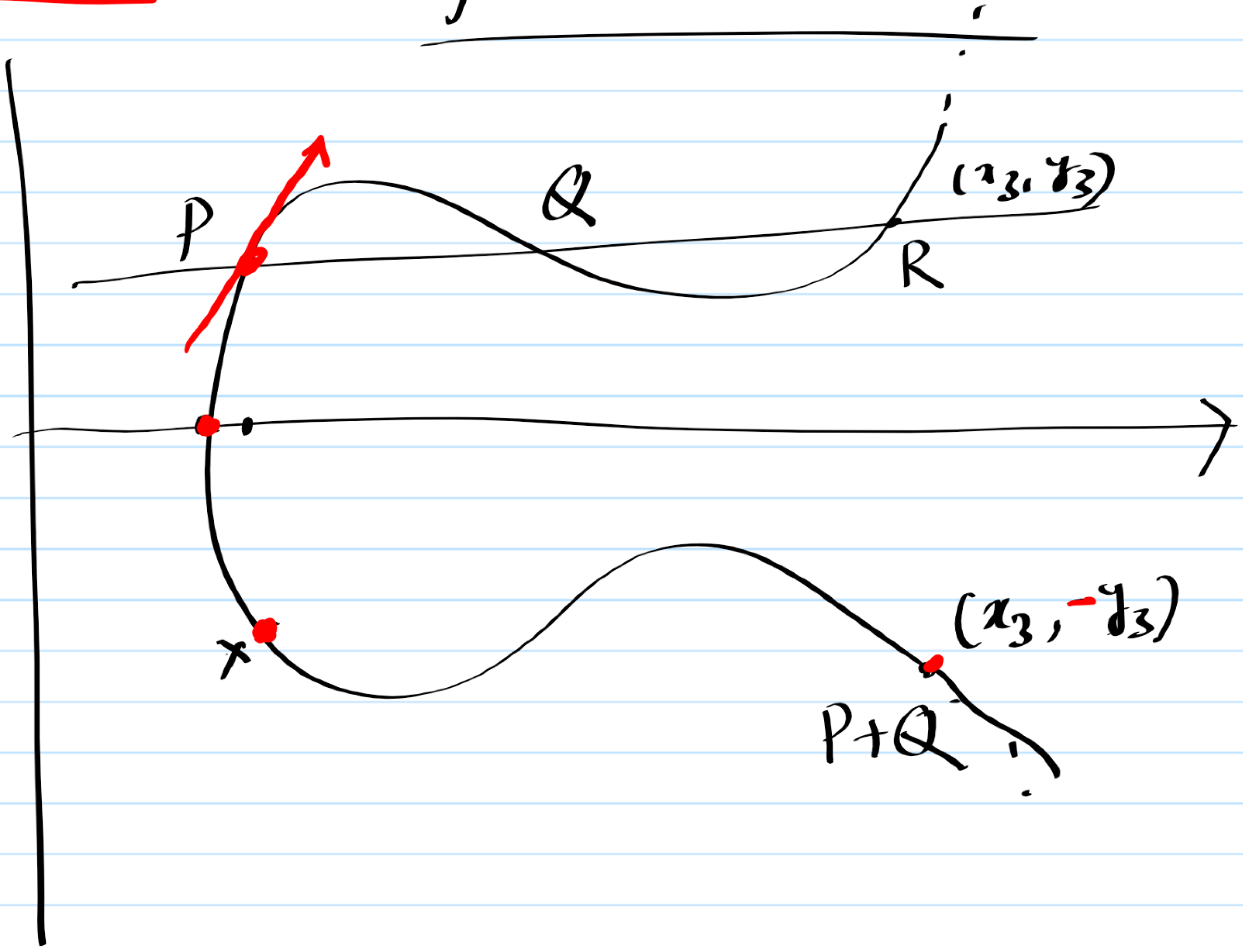
Char(F) = 0 or > 3

$y^2 = x^3 + ax + b$

$P = (x_1, y_1)$

$Q = (x_2, y_2)$

$\boxed{P \neq Q \ \& \ x_1 \neq x_2}$



$$\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$y = \left(\frac{y_2 - y_1}{x_2 - x_1}\right)(x - x_1) + y_1 \to (*)$$

$$\left[y = mx + y_1 - mx_1 \quad \text{where } m = \frac{y_2 - y_1}{x_2 - x_1} \right]$$

Substitute (*) in the eqn of the Elliptic curve

$$\{mx + y_1 - mx_1\}^2 = x^3 + ax + b$$

or $m^2x^2 + 2m(y_1 - mx_1)x + (y_1 - mx_1)^2 = x^3 + ax + b$

or $x^3 - m^2x^2 - \{2m(y_1 - mx_1) - a\}x - (y_1 - mx_1)^2 + b = 0$

$x_1 + x_2 + x_3 = m^2$ so $\boxed{x_3 = m^2 - x_1 - x_2}$ → (**)

$$\boxed{m = \frac{y_2 - y_1}{x_2 - x_1}}$$

$y_3 = mx_3 + y_1 - mx_1 = m(x_3 - x_1) + y_1$

$= m\{m^2 - x_1 - x_2\} + y_1 - mx_1$

$= m^3 - 2mx_1 - mx_2 + y_1$

$$\boxed{y_3 = m(x_3 - x_1) + y_1}$$

$y^2 = x^3 + ax + b$

Case $\boxed{P_1 = P_2, \ y_1 \neq 0}$

In this case $m =$ $\quad 2yy' = 3x^2 + a$

$$m = \frac{3x_1^2 + a}{2y_1}$$

at $P = (x_1, y_1)$

Eq^n of the tangent

$$y - y_1 = m(x - x_1)$$

$$y = m(x - x_1) + y_1$$

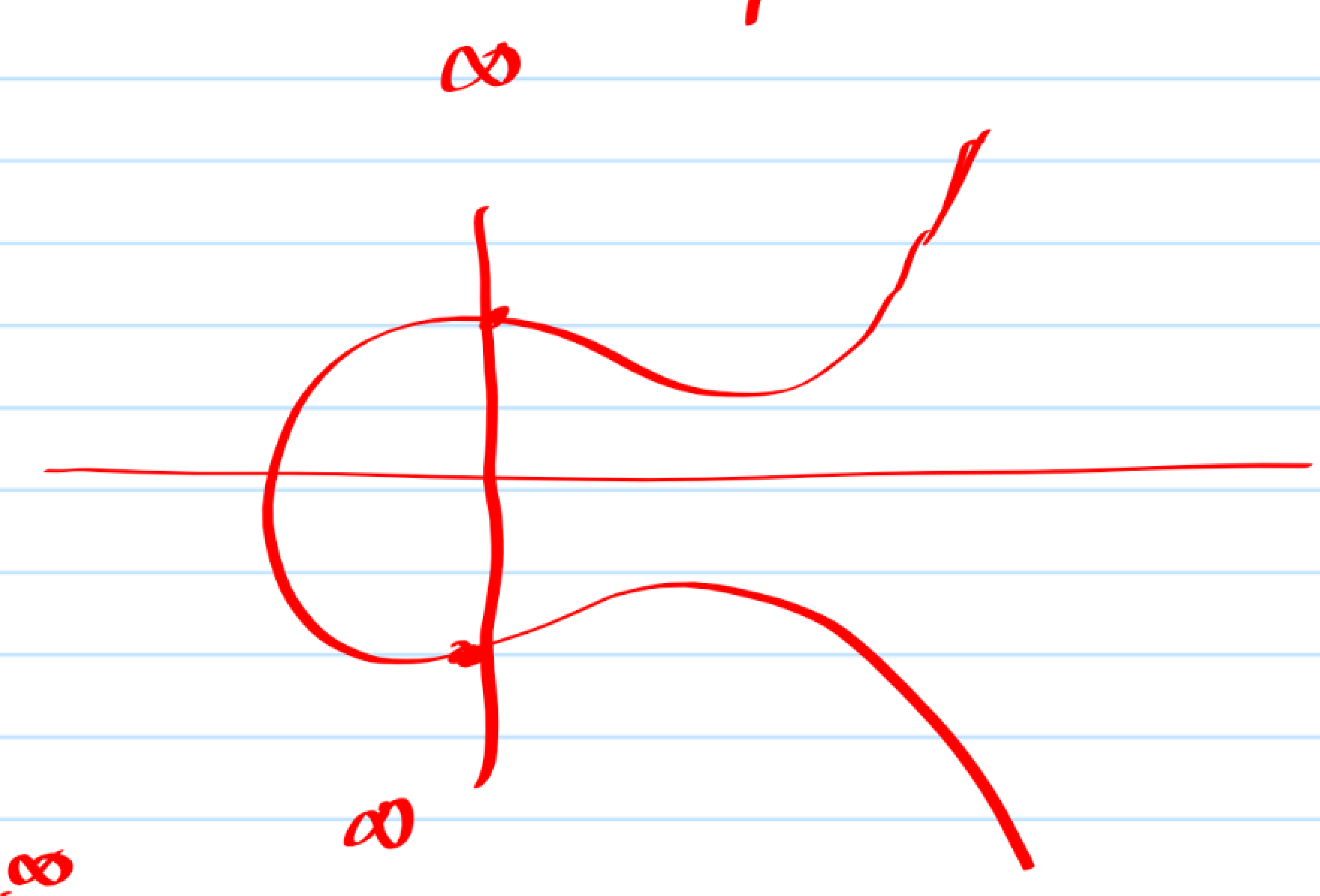
By substituting y in the eq^n of the elliptic curve we get $(**)$, a cubic in x.

$$x_1 + x_1 + x_3 = m^2$$

(since x_1, x_1, x_3 are the roots.

So $x_3 = m^2 - 2x_1$

$y_3 = m(x_3 - x_1) + y_1$ where $m = \dfrac{3x_1^2 + a}{y_1}$

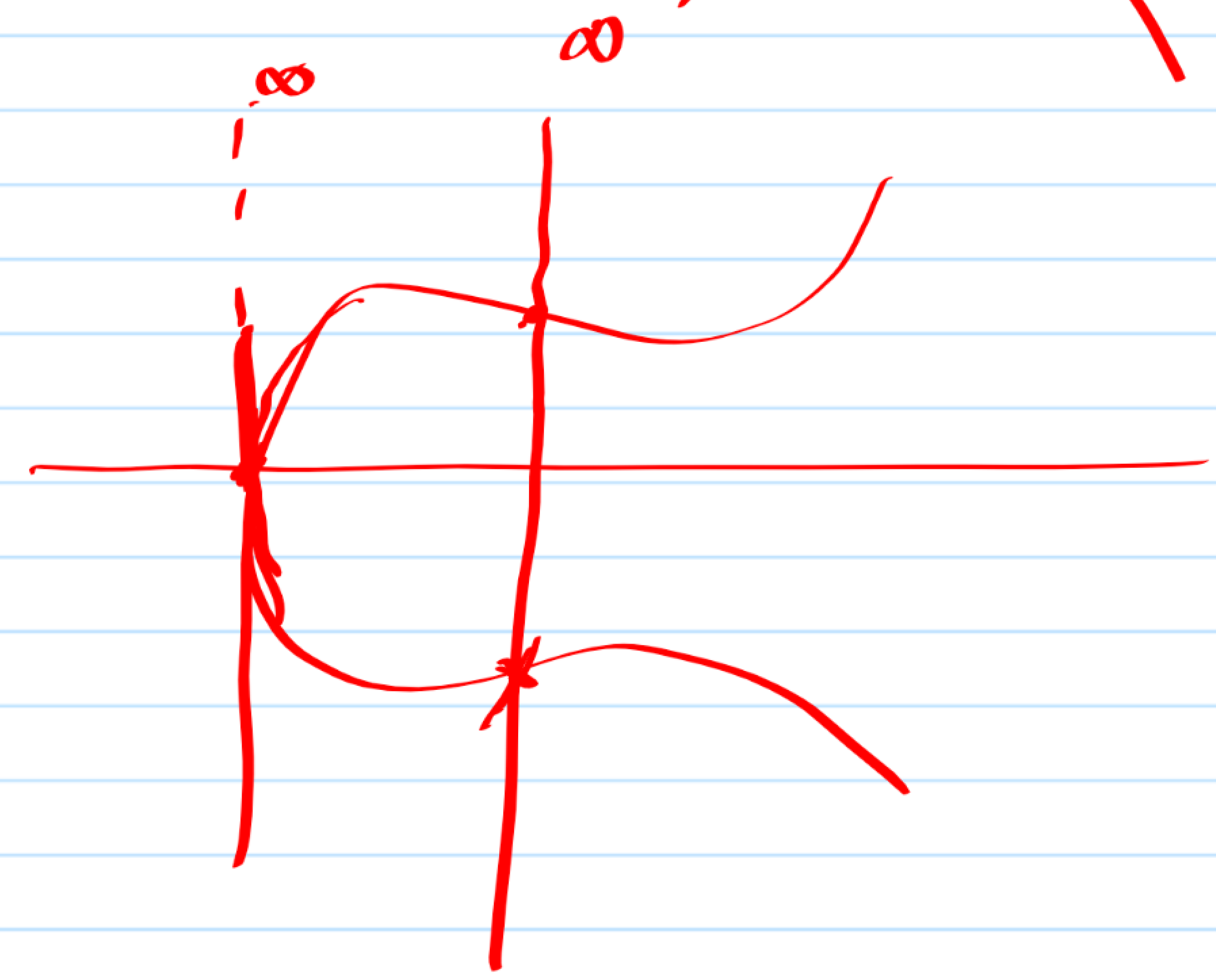


Case $x_1 = x_2 \quad y_1 \neq y_2$

$y_2 = -y_1$

$P_1 + P_2 = \infty$

$P_2 = -P_1$



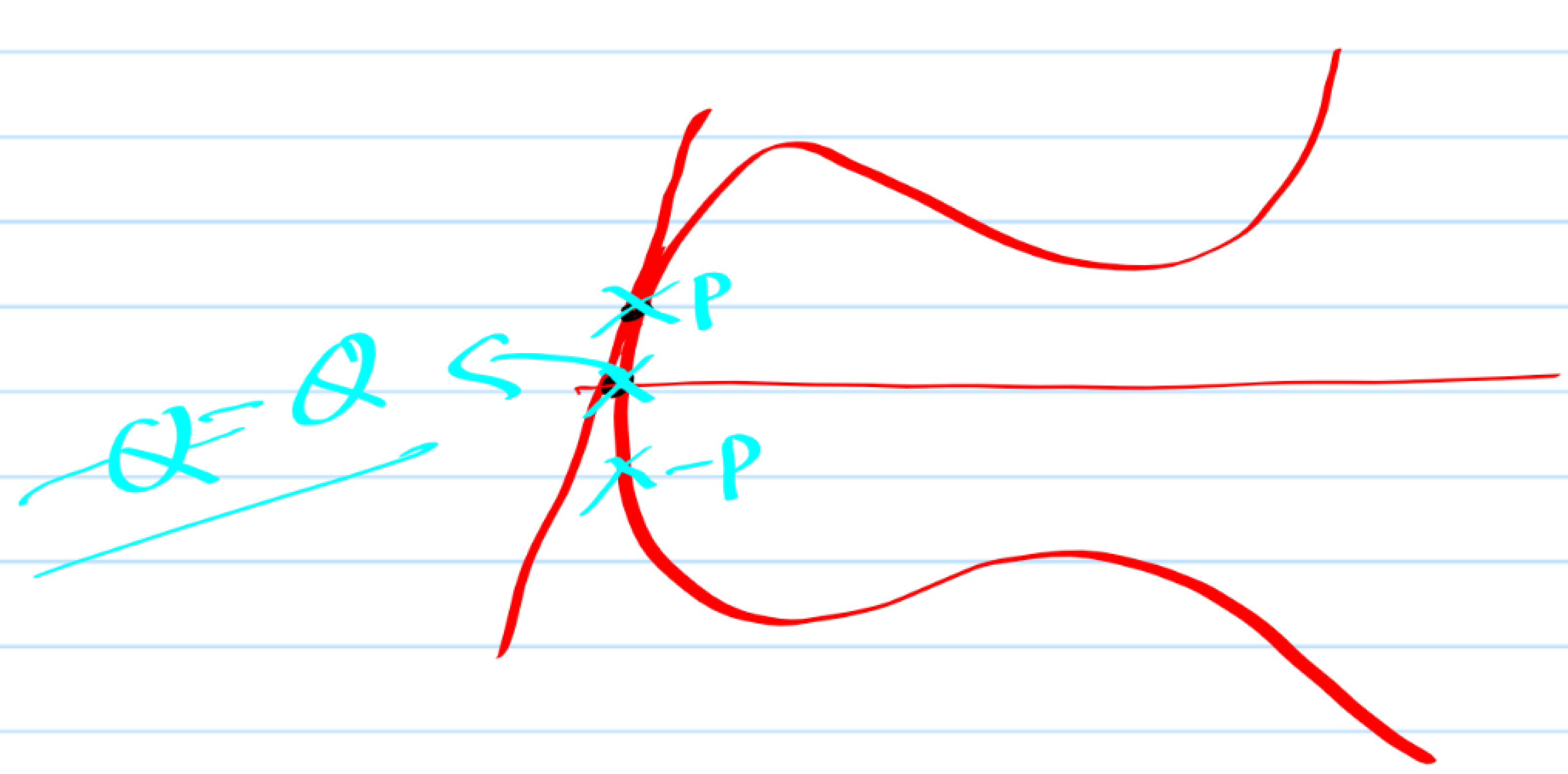
Case $P_1 = \infty, \; P_2 = (x_2, y_2)$

$P_1 + P_2 = +P_2$

$\therefore \; \infty + P_2 = P_2$

Case $P_1 = P_2 \quad y_1 = y_2 = 0$

$P_1 + P_2 = \infty$



Examples

$E = \{(x,y) \in \mathbb{Z}_{11}^2 : y^2 = x^3+3x+1\} \cup \{\infty\}$

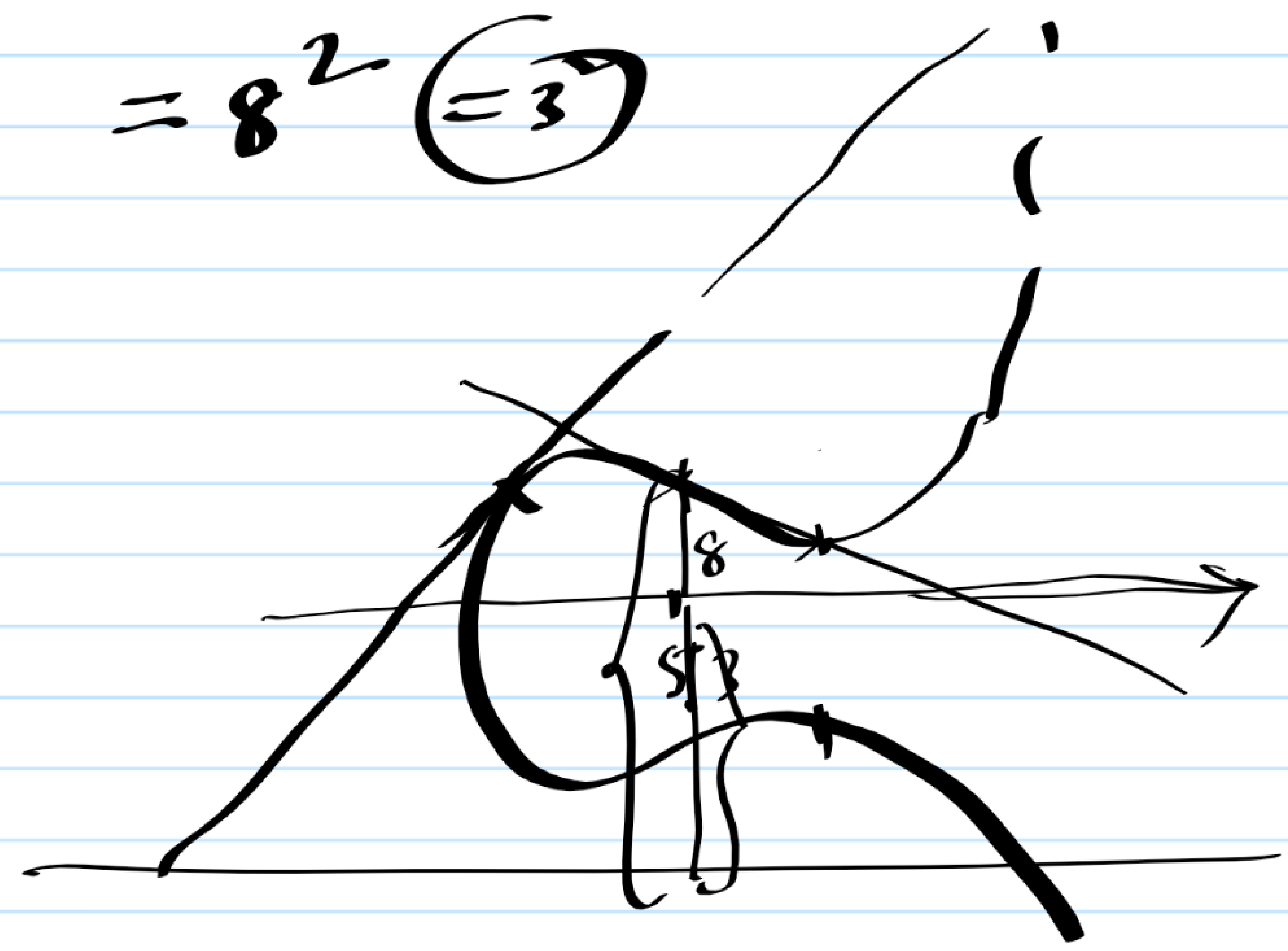
$P = (4,0) \in E.$

$4^3 + 3\times 4 + 1$
$= -2+1+1$
$= 0$

$2P = \infty$

$P = (5,8) \in E$

$5^3 + 3\times 5 + 1$
$= 4+4+1$
$= 9$
$= 8^2$ $=3$



$2P = ?$

$m = \frac{3x_1^2+a}{2y_1}$
$= \frac{3\times 5^2+3}{2\times 8}$
$= \frac{1}{5}$
$= 9$
$=$

$x_3 = m^2 - 2x_1$
$= 9^2 - 2\times 5$
$= 5$

$y_3 = m(x_3-x_1)+y_1$
$= 9(5-5)+8$
$= 8$

$2P = (5,-8) = (5,3).$

$3P = (5,3) + (5,8)$

$= \infty$

$m = \frac{y_2-y_1}{x_2-x_1}$

$\boxed{x_1 = x_2 \ \& \ y_1 \neq y_2}$

Consider $P = (0,1) \in E \equiv y^2 = x^3+3x+1$

$2P = (5,8)$
$3P = 2P+P = (4,0)$

$m = 7$
$x_3 = m^2 - 2x_1$
$= 7^2 - 2\times 0$
$= 5$

$2yy' = 3x^2+3$
$y' = \frac{3x^2+3}{2y}$
$= \frac{3}{2} = 6\times 3$
$= 7$

$m = \frac{y_2-y_1}{x_2-x_1}$
$=$

$y_3 = ?$

$$= \frac{8-1}{5-0} = 9 \times 7 = 8$$

$$x_3 = 8^2 - 5 - 0$$
$$= 9 - 5$$
$$= 4$$

$$y_3 = ? = 0$$

Verify:

$4P = (5,3)$

$5P = (0,10)$

$6P = \infty$

In this group (Elliptic curve) 6 is the order of P.

Verify:

$P = (1,4)$

$2P = (2,9)$, $3P = (0,1)$, $4P = (8,8)$, $5P = (3,9)$

$6P = (5,8)$, $7P = (6,2)$, $8P = (9,8)$, $9P = (4,0)$

$10P = (9,3)$, $11P = (6,9)$, $12P = (5,3)$, $13P = (3,2)$.

$14P = (8,3)$, $15P = (0,10)$, $16P = (2,2)$ $17P = (1,7)$

$18P = \infty$.

Quiz on 24th